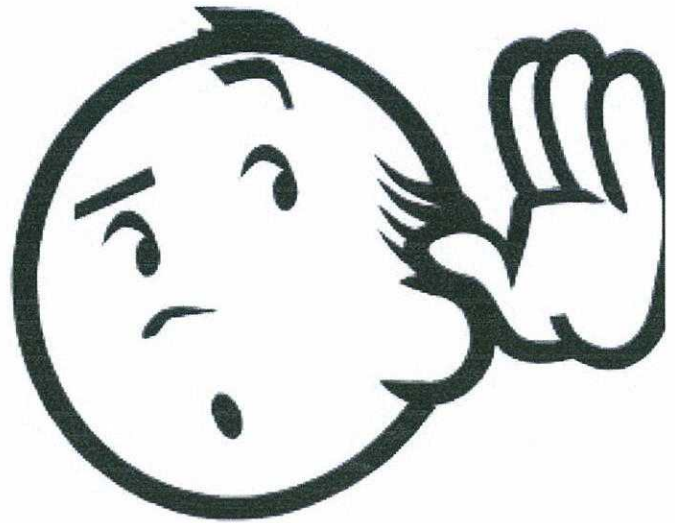
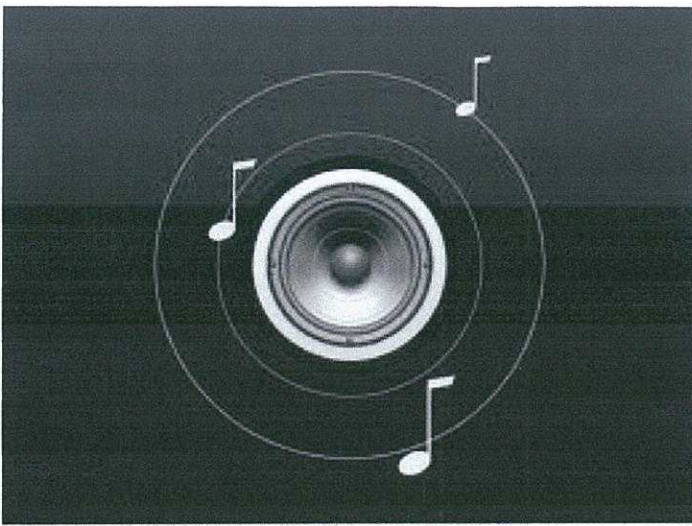


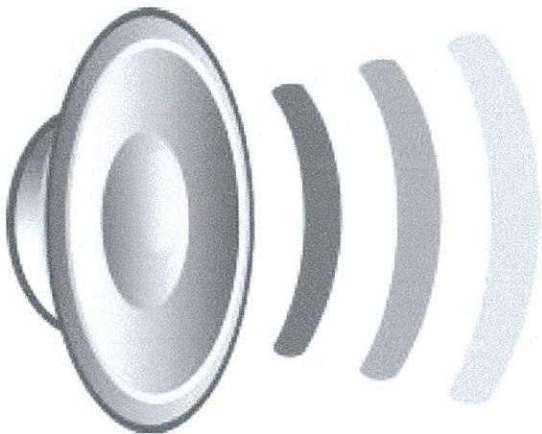


YEAR 10 SCIENCE

ENERGY AND CHANGE

SOUND

Name: _____



WAVE MOTION

Waves transmit energy from one point to another in a substance (called a medium) without the particles of the substance moving with it.

This leads to the following definition.

Wave motion: a disturbance or vibration that transmits energy without the movement of matter from one point to another.

Waves that require a medium are called **mechanical waves**.

e.g. Sound, water waves, waves in strings and springs.

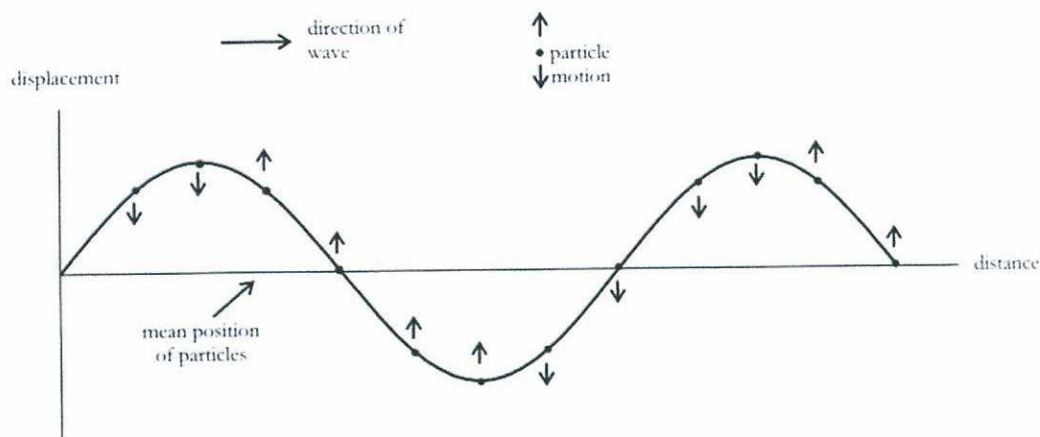
Waves that don't need a medium are called **electromagnetic waves**.

e.g. Light.

There are two types of mechanical waves.

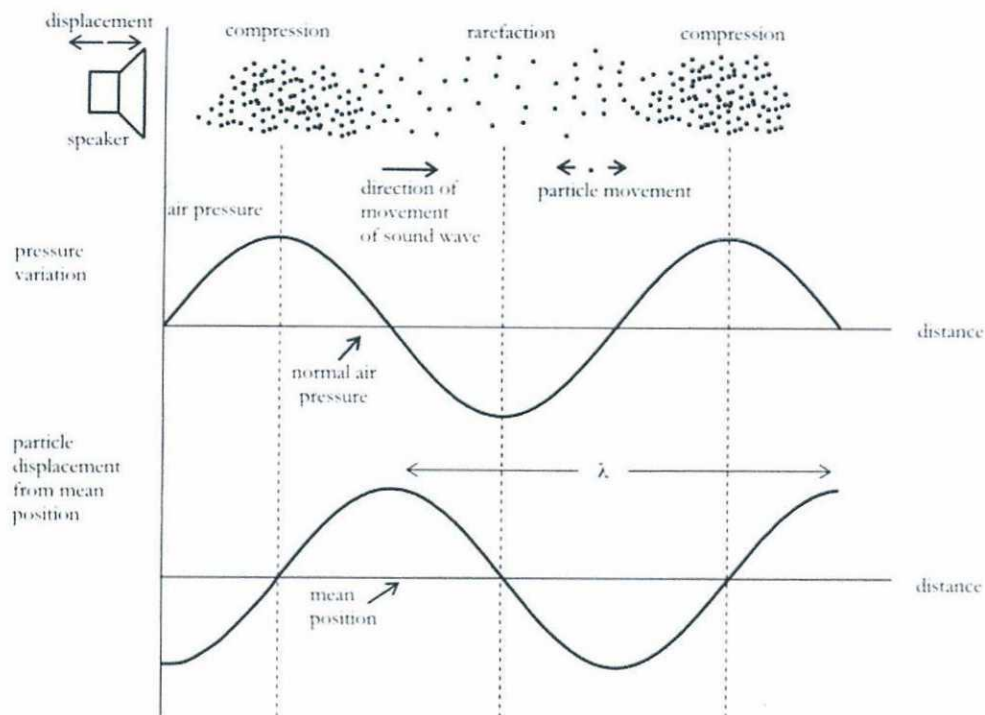
Transverse wave - particles move **perpendicular (up and down)** at right angles to the direction of the movement of the wave.

e.g. Water waves.



Longitudinal wave - particles move *parallel (back and forward)* in the same direction as the direction of the movement of the wave.

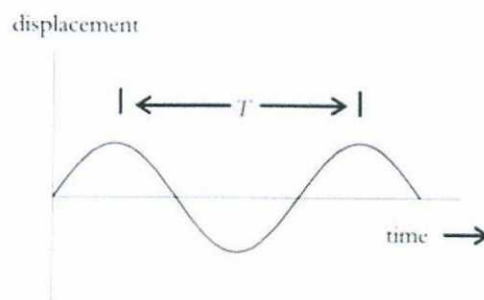
e.g. Sound waves



Wave graphs

Wave motion can be represented by a displacement/ time graph or a displacement/distance graph.

- Displacement / Time**

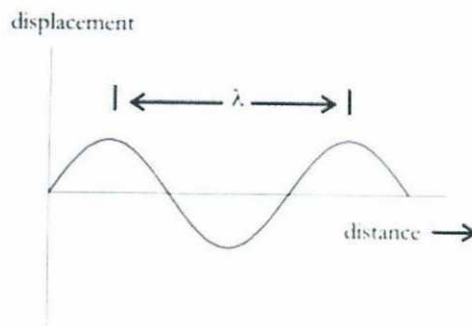


This graph represents the *motion of a single particle* as a wave passes through it.

The crests and troughs represent the particle's maximum displacement from its mean position.

T is the period of vibration - time taken for one complete oscillation (up-down and back to start position).

- **Displacement / Distance**

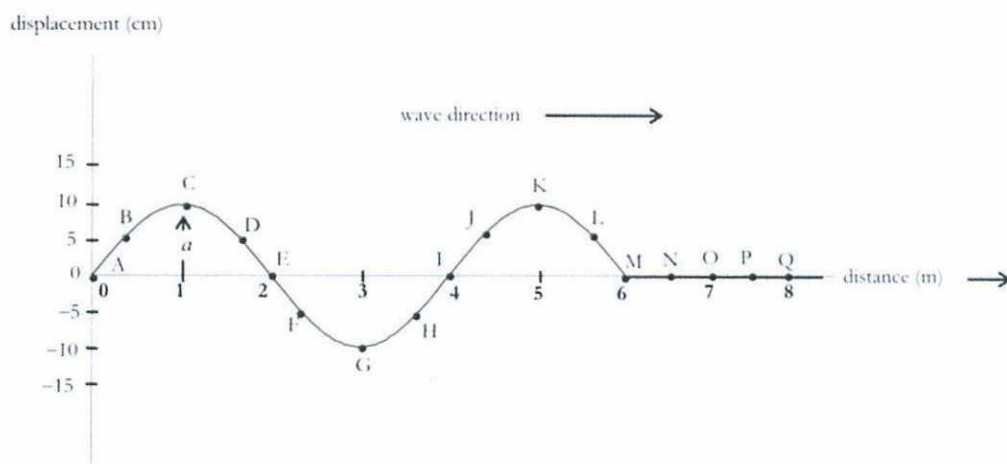


This graph represents the *movement of a number of particles* from their mean position at some instant in time.

Think of it as a "snapshot" or photograph that shows how the displacement of the particles of the medium varies over the distance at that instant.

λ represents the *wavelength of the wave*.

Describing Wave Motion - Important Terms



Wavelength (λ) The distance between any two crests or troughs.

e.g. The distance between points C & K, A & I, etc.

Amplitude (a) The maximum displacement of a particle from its mean (normal) position.

i.e. Height of a crest or depth of a trough.

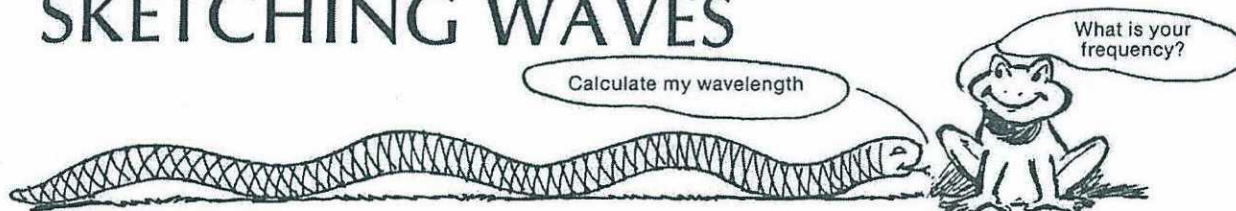
The amplitude of the wave shown is 10 cm.

Frequency (f) The number of complete vibrations (or waves) per second.

The unit of frequency is the hertz (Hz).

Period (T) The time taken to complete one vibration (or to produce one complete wave).

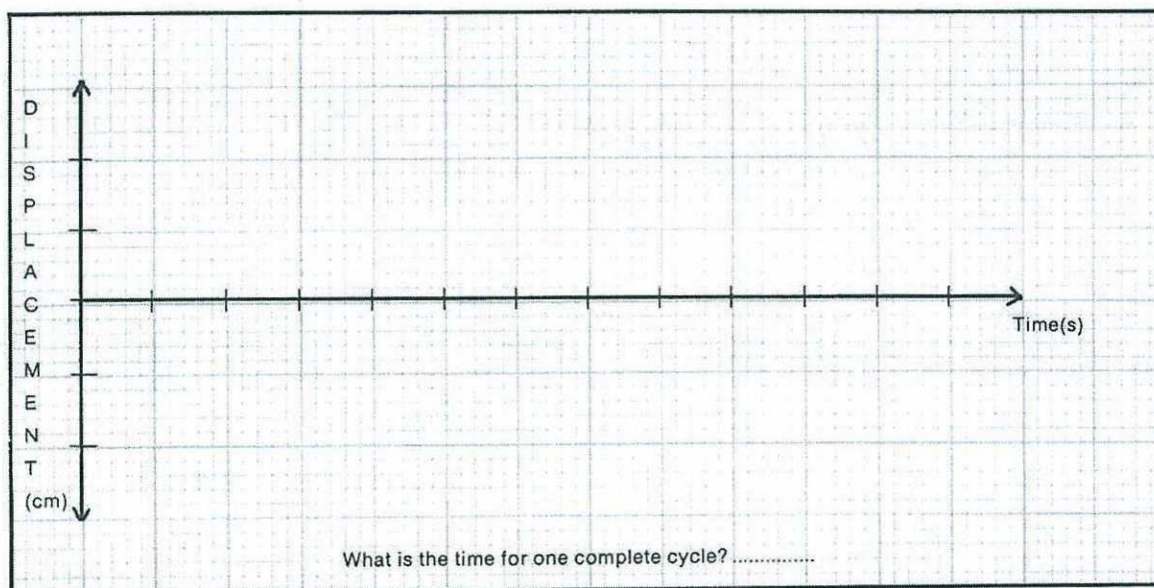
SKETCHING WAVES



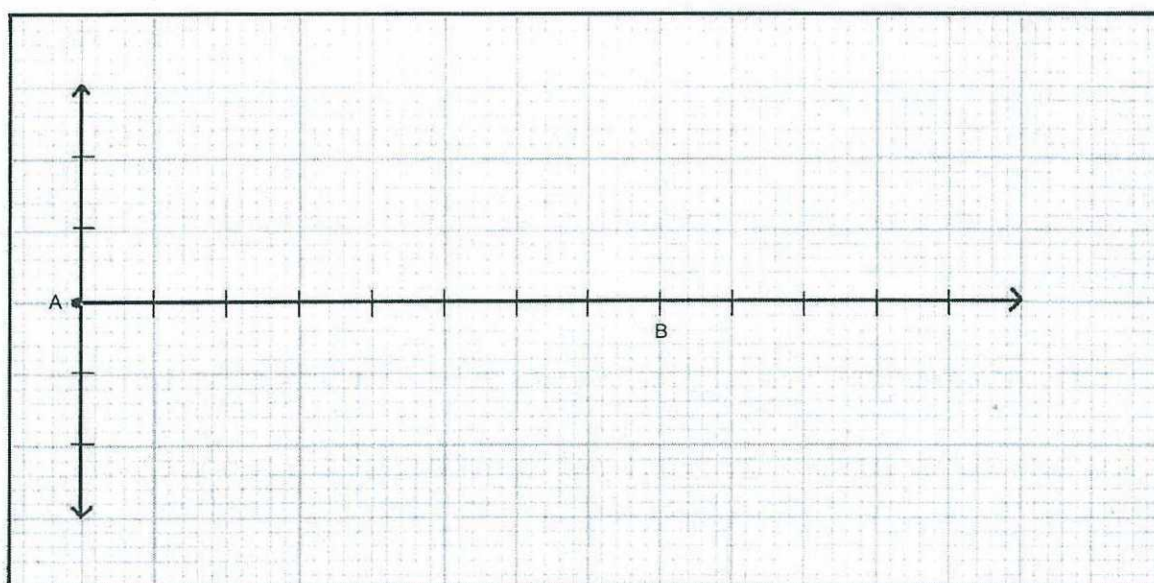
The following data have been obtained by photo-analysis of the motion of a single particle in a medium as energy in a wave form passes through the medium.

Time (s)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
Displacement from rest position (cm)	0	5	11	16	19	20	19	16	11	5	0	-5	-11	-16	-19	-20	-19	-16	-11	-5	0

Plot the data on the graph paper.



Using the piece of wire you have been given, construct the outline of a wave with a wavelength of 4 cm and an amplitude of 3 cm. Trace the outline onto the graph paper.



If the wave takes 1 second to travel from A to B, calculate the frequency.

SOUND AND WAVES

Background

Sound is a form of wave motion that we use every day. What would our world be like if it were silent? Wave motion has various characteristics that need to be understood and learned.

Task

Your task is to use the Internet to find useful information that can provide *simple answers* to the following questions. Each set of questions has a website that should be visited first.

You may find others by using Metacrawler (<http://www.metacrawler.com/>) and doing a global search on waves, sound, or something similar. Applets are interactive and may be very useful.

Questions

Go to the following website.

<http://www.ngsir.netfirms.com/englishVersion.htm>

- Click on Light and Wave.
- Select Transverse Wave (run).
- Select Start.
- Choose the Select a Point button.

1. How does the selected point move?
2. What happens when you change the following?
 - (i) Amplitude
 - (ii) Wavelength
 - (iii) Speed

- Go back one page.
- Select Longitudinal Travelling Wave (run).
- Choose the Select a Point button.

3. How does the selected point move?
 - Select the Graph button - the movement now shows as a graph.
4. Draw what the shape of the graph looks like.

5. What happens when you change the following?

- (i) Amplitude
- (ii) Wavelength
- (iii) Speed

Go to the following website.

<http://www.surendranath.org/Applets.html>

- Select Applet Menu
- Choose Waves.
- Select Transverse Waves.

6. What happens when you change the amplitude and frequency?

- Repeat for Longitudinal Waves.

Go to the following website.

<http://www.fi.edu/fellows/fellow2/apr99/soundindex.html>

- Click on How Sound is Made.

7. What are sound waves?

8. In which direction do the particles of the medium move as a sound moves through it?

9. Draw a simple diagram to show how a tuning fork produces a sound.

10. What is a:

- (i) compression?
- (ii) rarefaction?

11. Can a sound wave travel through a vacuum? Briefly explain your answer.

Go back 1 page.

- Click on Music to Our Ears.
- Click on The Ear.

12. What do the following parts of the ear do?

- (i) Outer ear.
- (ii) Ear drum.
- (iii) Hammer, anvil and stirrup (in the Middle Ear).
- (iv) Cochlea.
- (v) Auditory nerve.

13. Do sound waves enter our brain? What does?

Go back 2 pages.

- Click on Good Vibrations.

14. On what device can we see a picture of a sound wave?

15. Define the following.

- (i) Wavelength
- (ii) Frequency
- (iii) Amplitude

16. Draw a graph to show the difference between the following.

- (i) A high pitch and a low pitch sound.

- (ii) A loud and soft sound.

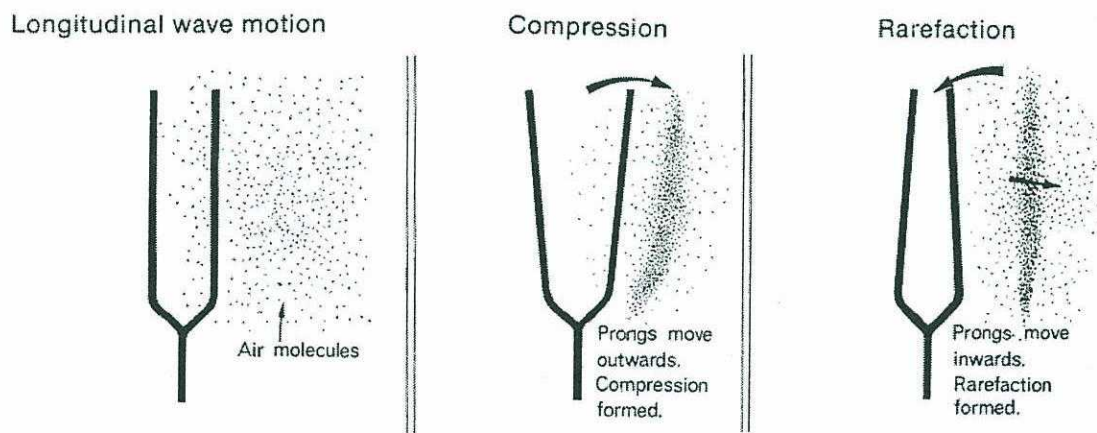
- (iii) A pleasant sound and noise.

PRODUCTION OF SOUND

Sound waves are produced by a **vibrating source**.

They need a **medium** to travel in, such as air, water or a solid. Sound cannot be heard in space.

e.g. A tuning fork.



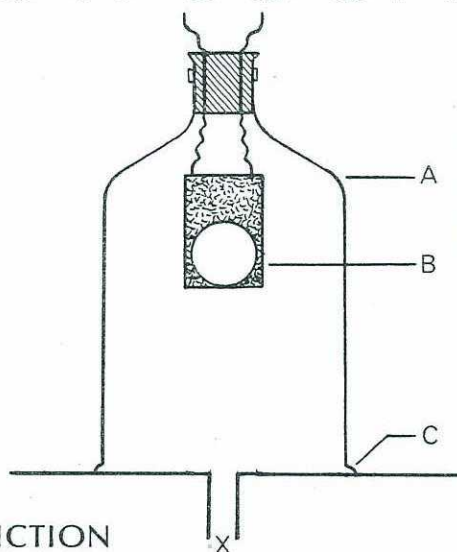
The sound waves produced are **longitudinal waves** made up of a series of **compressions** (high pressure areas) and **rarefactions** (low pressure areas).

Energy is transferred through the medium without the particles moving from one place to another.

The **frequency of vibration** determines the sound that is heard - a high frequency sounds high, a low frequency sounds low.

e.g. Jet engines have a high frequency whine while ceiling fans have a low hum.

HOW SOUND TRAVELS



Examine the diagram and, after some discussion with other members of your group, predict what will happen when the bell is switched on and the system is connected to a vacuum pump. You may like to support your prediction with some reasons.

- A Bell jar
- B Electric buzzer
- C Petroleum jelly seal
- X To vacuum pump

GROUP PREDICTION

REASON

Your teacher has set up some similar equipment which will allow the class to observe what actually happens.

OBSERVATIONS

CONCLUSIONS

1. Did the results from the demonstration support your group's prediction?

2. Can you suggest an explanation for the experimental results?

3. Make some suggestions as to how this experiment could be improved.

4. From this activity, can you suggest what is needed before sound can be heard?

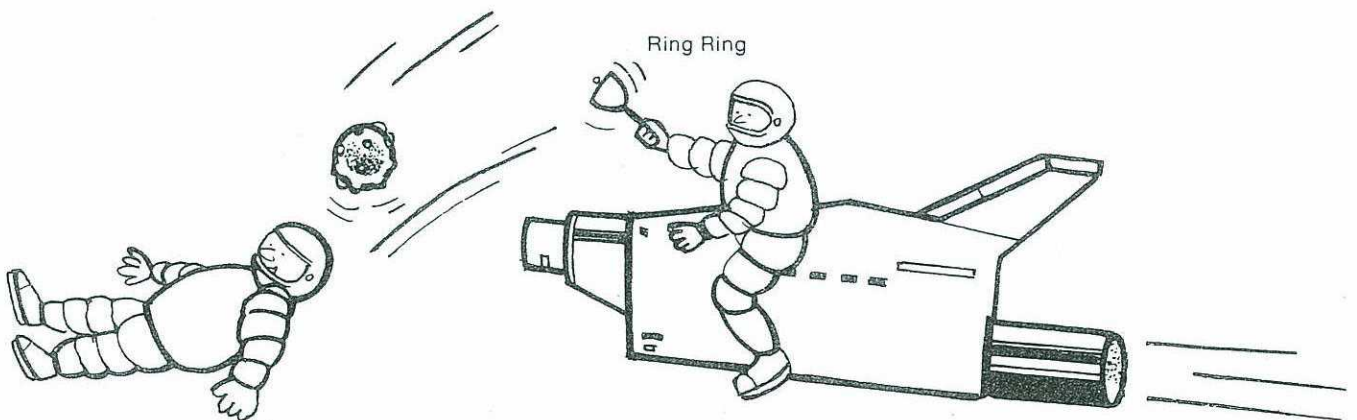
MORE NOISE

DISCUSS these diagrams and try to work out what is wrong.



SPOT THE ERROR

This cannot happen because _____



SPOT THE ERROR

This cannot happen because _____

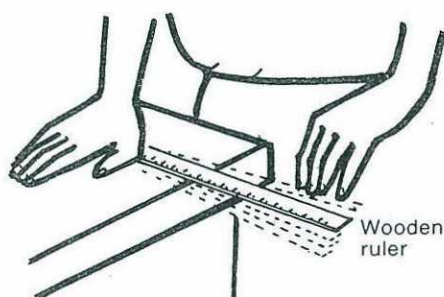
PRODUCE THE FOLLOWING SOUND



PROCEDURE

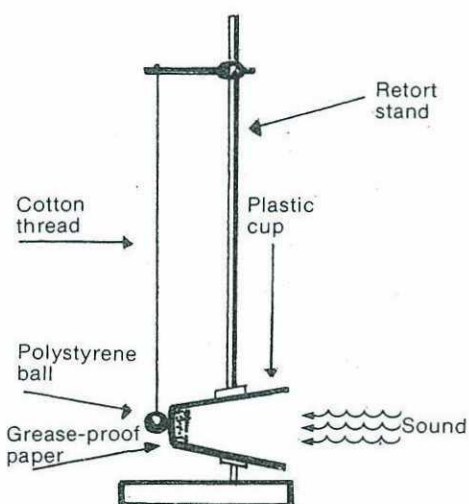
Cut the bottom out of a plastic cup, then stretch some adhesive food wrapping over the cut end of the cup. Speak into the open end of the cup and record your observations.

OBSERVATIONS: _____



Place a ruler on the top of the bench and have 15 to 20 cm overhanging the edge. Holding the ruler firmly on the desk with one hand, pluck the free end and record your observations.

OBSERVATIONS: _____

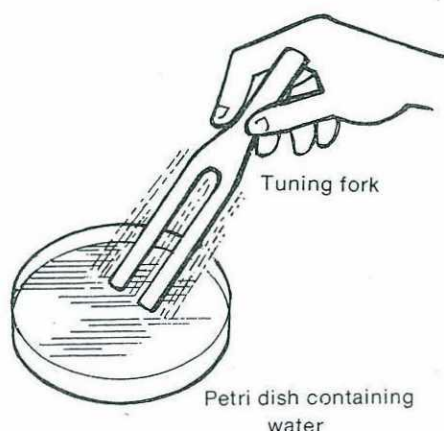


Set up the equipment as shown in the diagram, with the small polystyrene ball resting against the centre of the paper. (*Note: the thread is not quite vertical.*) Before you speak into the cup, make a prediction.

I predict _____

Test your prediction.

Explain the result. _____



Sound a tuning fork by striking the prongs on a *soft* surface. While the fork is vibrating, gently touch the prongs on the surface of the water. Record your observations.

OBSERVATIONS: _____

Can you explain your observations? _____

BRIEFLY SUMMARIZE YOUR OBSERVATIONS _____

HOW SOUND TRAVELS THROUGH A GAS

The incomplete diagrams below show how sound travels through a gas. Copy the diagrams into your notes. When you have done this, complete the diagrams you have copied. You will need to use a textbook to find out more about how sound travels through a gas.

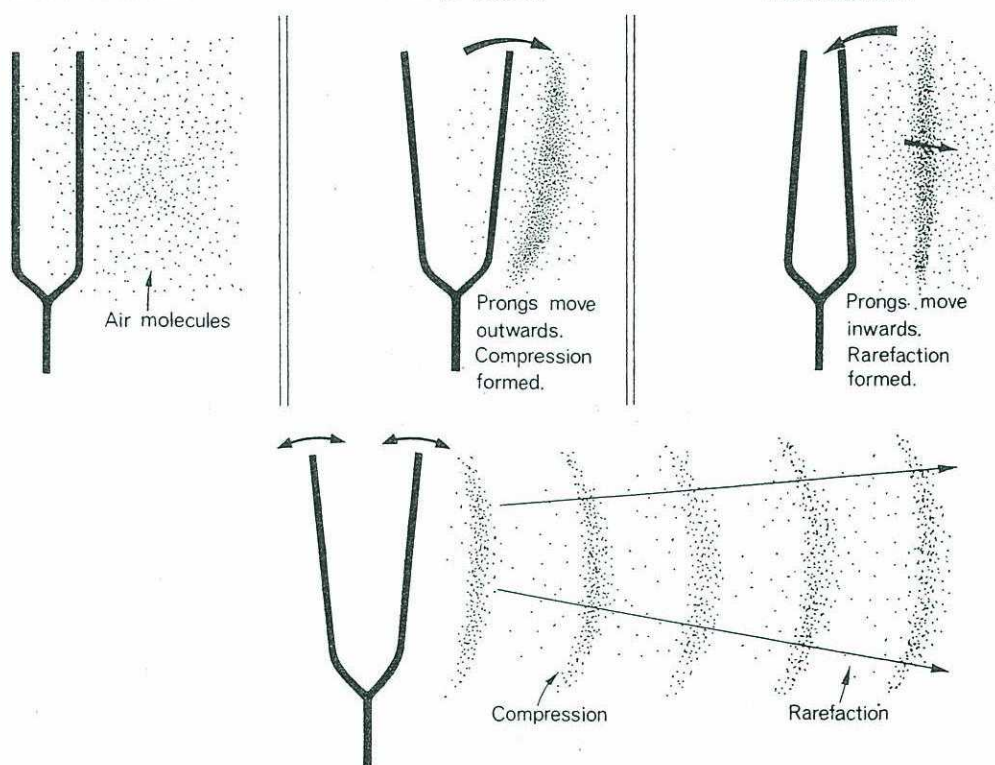
To help, some key words have been listed.

KEY WORDS

Longitudinal wave motion

Compression

Rarefaction



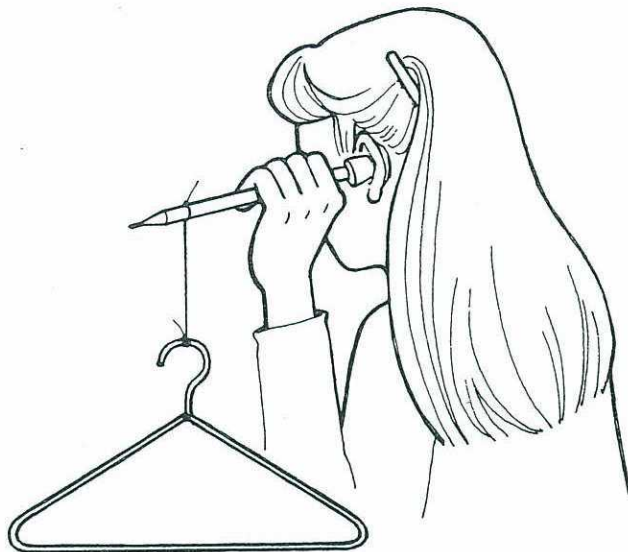
Complete the diagrams, showing what happens on the other side of the tuning fork. You must be sure to label the compressions and rarefactions formed. Beneath the diagrams write a brief paragraph describing, in your own words, how sound travels through a gas.

BRAIN TEASERS

TRY THESE!

1. A gas tap is opened on the front desk. The sound of the escaping gas is heard almost instantly at all parts of the room. After some time, the smell of the gas is detected. What does this show?
2. Explain why a sound becomes softer as you move away from its source.
3. Why is it difficult to hear or to make yourself heard when shouting into the wind?
4. Sound can be heard over longer distances after it has been raining. Why do you think this occurs?

SOUNDS IN SOLIDS



Note—make sure the pencil does not go all the way through the stopper.

PROCEDURE

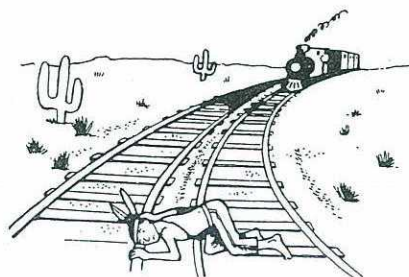
1. Carefully place the stoppered end of the pencil to your ear.
2. Hit the coat hanger with another pencil.

RESULTS

1. What do you hear with the ear to which the stoppered end of the pencil is placed?
2. What was the source of the sound?
3. Why didn't you hear the coat hanger vibrations through the air?
4. Describe how the vibrations reached the ear from the source.

EXPLAIN THESE

1. A 'knock' in a car engine can be located by putting one end of an iron rod here and there on the engine block and holding the other end near your ear.
2. If an object were struck under water, would a nearby skindiver hear a sound?
3. Explain how people trapped in a mine cave-in could be located.
4. Write a brief description of what is happening in the diagram.



SPEED OF SOUND

Sound travels fastest in a solid, slowest in a gas.

The particles are much closer together in a solid so they collide with each other very quickly so the sound energy is transferred much faster than in a liquid or gas.

MEDIUM	SPEED OF SOUND
air	340 ms^{-1}
water	~ 4 times faster than air
solid	~ 15 times faster than air

To calculate the speed of sound:

$$\text{Speed of sound} = \frac{\text{distance travelled}}{\text{time taken}}$$

EXAMPLE

The sound of thunder is heard 4.8 seconds after the flash of lightning. If the velocity of sound is 340 ms^{-1} , how far was the lightning away from the observer?

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{d}{t}$$

$$\Rightarrow 340 = \frac{d}{4.8}$$

$$\begin{aligned}\Rightarrow d &= (340)(4.8) \\ &= 1632 \text{ m}\end{aligned}$$

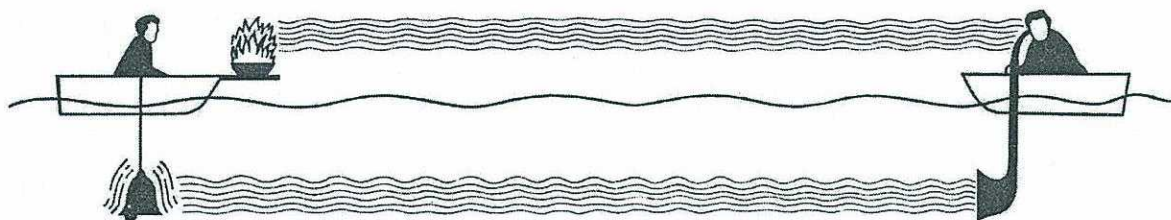
$\therefore$ The lightning was 1632 m away.

HOW CAN YOU MEASURE THE SPEED OF SOUND?



1. What differences between the properties of sound and light are used in this experiment?
2. Why did this experiment make the person think of thunder and lightning?

The Colladon Experiment



In 1826, a historic experiment was carried out on Lake Geneva. Two boats were moored 13 500 metres apart. One carried a 70 kg bell (below the water) and a small tray of gunpowder.

The gunpowder was ignited just as a hammer struck the bell. An observer in the second boat was able to see the flash from the gun-powder and, by using an ear trumpet, to hear the sound of the bell. He started his stop-watch when he saw the flash and stopped it when he heard the sound in the ear trumpet.

1. What was being measured in this experiment? Do you think the results would be accurate?
2. How did the distance between the boats affect the time it took for the listener to hear the explosion? Did this affect the result of the experiment? Explain.
3. The observer heard two explosions. Explain why this happened.
4. What effect would atmospheric conditions, such as wind speed and direction or air temperature, have had on the accuracy of the result? Explain.
5. Explain how the following affected the experimental results:
 - (a) The observer's reaction time.
 - (b) The depth of the bell in the water.
 - (c) The purity of the water.
 - (d) The temperature of the water.
6. The scientists repeated the experiment many times before they published their results. Explain why they collected so many results.

REFLECTION OF SOUND

The amount of sound reflected from a surface is dependent on the nature of the surface.

A hard surface (e.g. brick) reflects most sound energy (a little is absorbed) while a soft surface (e.g. carpet, leather) absorbs most sound energy (a little is reflected).

DEFINITIONS

ECHO: reflected sound heard separately from the original sound.

To hear an echo, the two sounds must be at least 0.1 seconds apart.

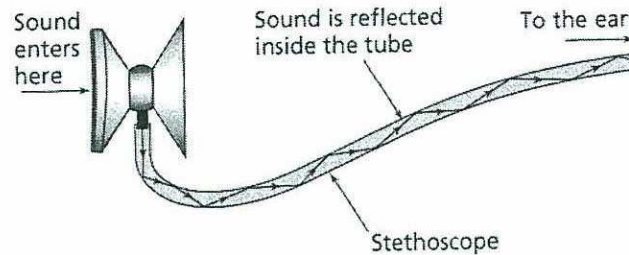
REVERBERATION: multiple echoing that causes distortion of sound.

This occurs in large empty halls where there is an absence of sound-absorbing materials.

APPLICATIONS

Stethoscope

- The sound of a patient's heart or breathing is reflected up the tubes to the doctor's ears.



Music bowl or sound shell

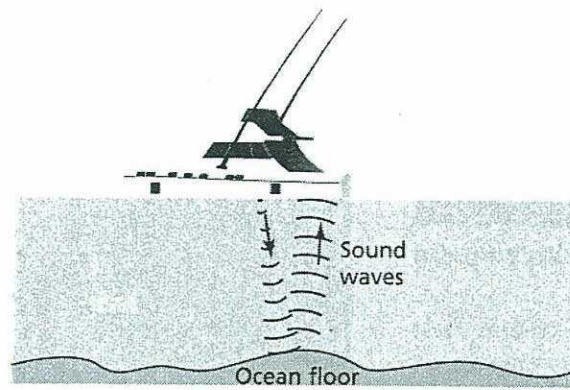
- Reflects the sound towards the audience out front.

Ultrasound

- Reflects sound waves from parts inside the body.
- Computer analyses reflections to produce an image.

Sonar or depth sounding

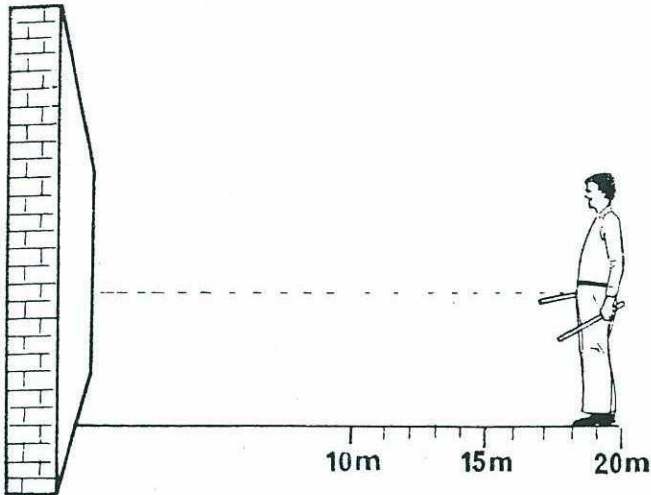
- Sound is bounced off the ocean floor to give the depth.
- Submarines bounce sounds off ships to detect them.
- Bats bounce sound off insects to find them in the dark.



Acoustics

- Sound auditoriums, concert halls and recording studios are designed to minimise reflected sounds.
- Soft furnishings, curtains, people all act as sound absorbers.
- Balconies are staggered to reflect "unwanted sound" away from the audience.
- Reverberation can be an advantage - it adds "richness" or "depth" to music played in an auditorium. Bands playing outside with no reverberation sound "thin" or "tinny".

ECHO



- Measure a distance of 10 metres from the wall. Mark that spot.
- Mark every metre for the next 10 metres.
- Stand 10 metres from the wall and hit the two blocks together. Record your observation.
- At one metre intervals, hit the blocks together and record your observation.

DISTANCE FROM THE WALL (m)	OBSERVATION
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	

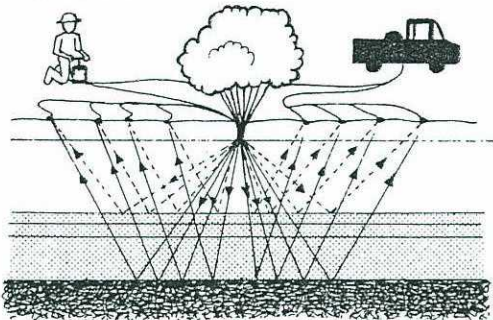
- How far from the wall were you when you first heard the reflected sound?
- Explain why no echo was heard until you reached a certain distance from the wall.
- How could you use the data from this experiment to work out the speed of sound?

ECHOES AT WORK

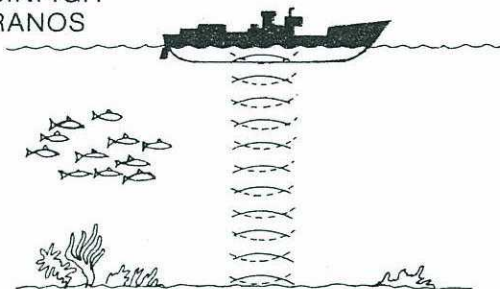
UNSCRAMBLE THE WORDS

Use the pictures to give you clues which can help you unscramble the words. When you have unscrambled the words, find out what they mean, and briefly describe how they apply to the particular picture.

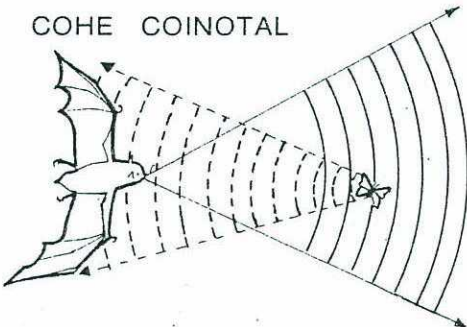
CESISIM



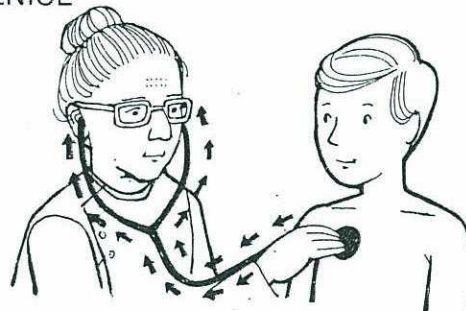
SINFIGH
RANOS



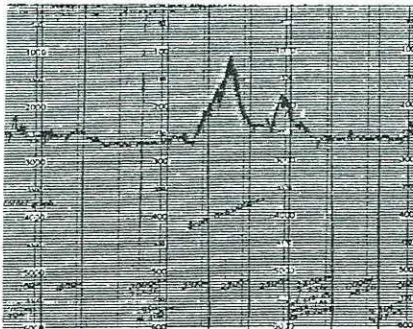
COHE COINOTAL



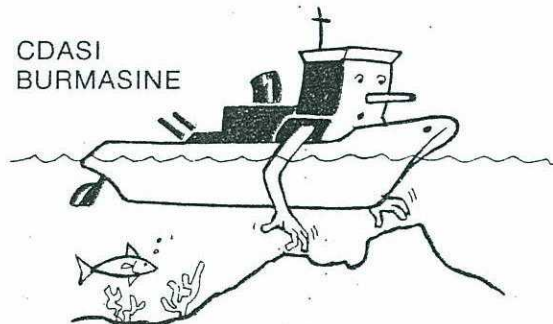
DIMENICE



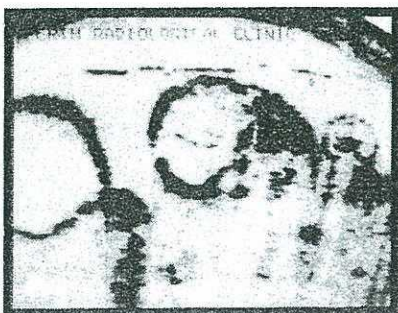
COHE NOGINSUD



CDASI
BURMASINE



SWITN LURODSANUT



COHE COINOTAL



RESONANCE

RESONANCE: where an applied frequency (driving frequency) matches the natural frequency of a body, causing a large increase in amplitude of the vibrations.

- e.g. 1. Blowing over the top of a bottle causes the air column inside to resonate and produce a sound.

This is the basic principle of all wind instruments.

2. At certain speeds, loose parts on old cars start to resonate due to the vibration of the motor or road.
3. People walking over suspension bridges in step will cause the bridge to resonate to their steps. A large amplitude in vibration can result which ultimately can destroy the bridge.

FORCED VIBRATION: where an applied frequency forces an object to vibrate at a frequency other than its natural frequency.

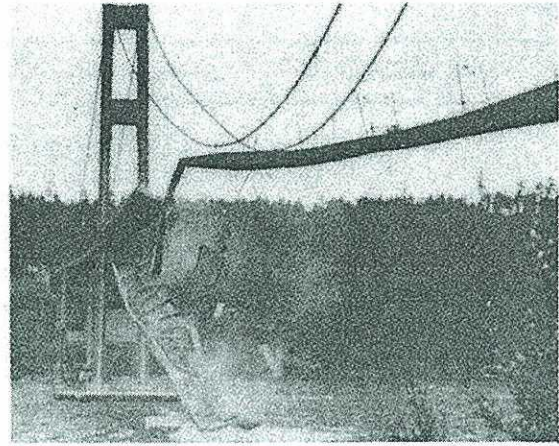
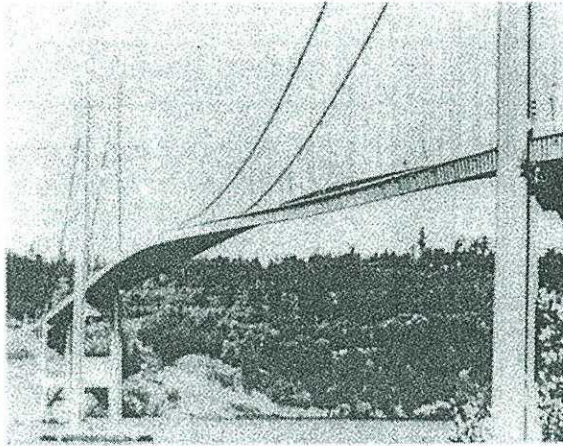
- e.g. A vibrating tuning fork in contact with a table causes the table to vibrate at the same frequency.

Since the area of the table in contact with the air is greater, an increase in the amplitude of the sound is heard.

MORE RESONANCE

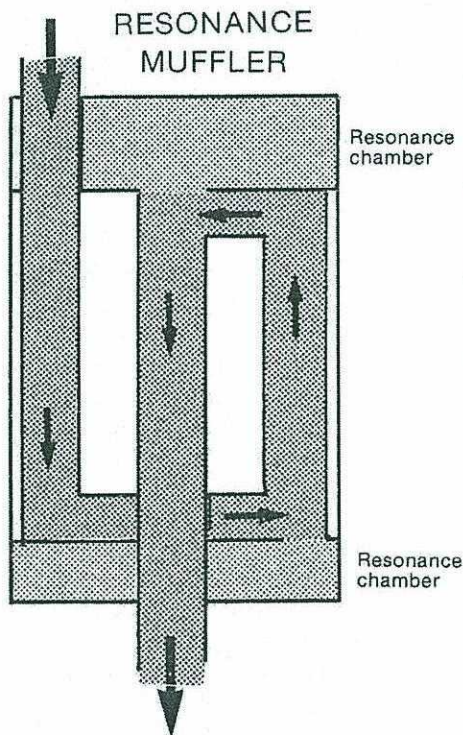
WHAT HAPPENED HERE?

THE TACOMA BRIDGE DISASTER



1. Explain how a gusting wind caused this bridge to collapse.
2. Why wasn't the strength of the wind the most important factor in this disaster?
3. What special feature of this bridge may have made it vulnerable to this type of failure?

HOW DOES IT WORK?



1. How is it possible for a resonance chamber to make a noise quieter?
2. Explain why the resonance chambers are of different sizes.
3. What does the loudness of the sound indicate?
4. Draw a flow diagram, showing the energy exchanges taking place in the system.

TYPES OF SOUNDS

NOISE: made by a source whose vibrations **have no pattern** (irregular vibrations).

MUSIC: made by a source whose vibrations **have a pattern** (regular vibrations).

Noises tend to be annoying. Loud noises can cause temporary or permanent hearing loss (so can loud music).

Hearing protection equipment (ear muffs, ear plugs) are important for people exposed to loud noise for long periods of time.

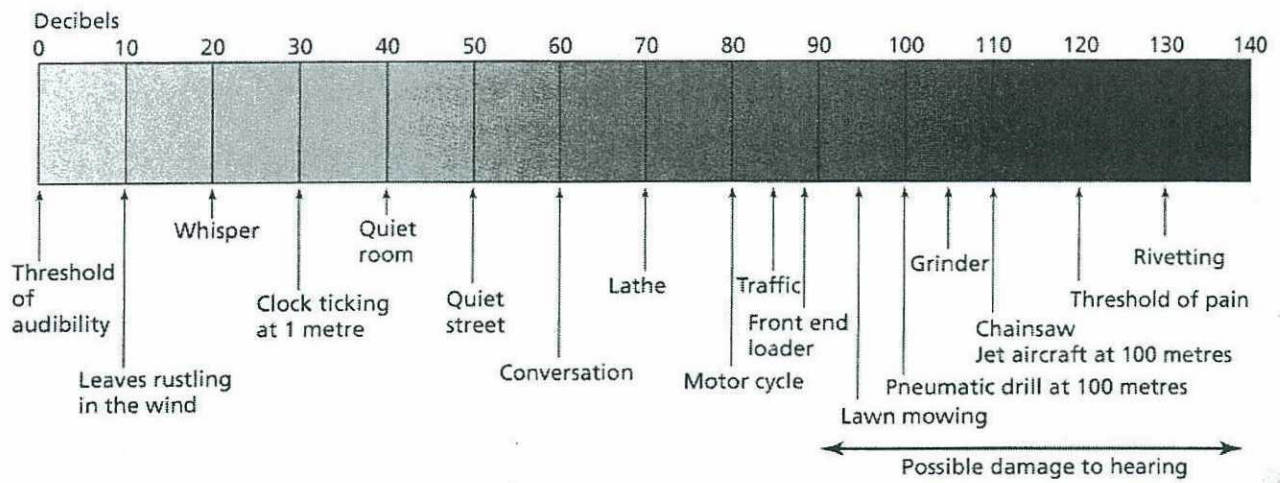
Sounds are measured in units called **decibels (dB)**.

Normal conversation is about 60 dB.

Yelling in someone's ear is about 110 dB.

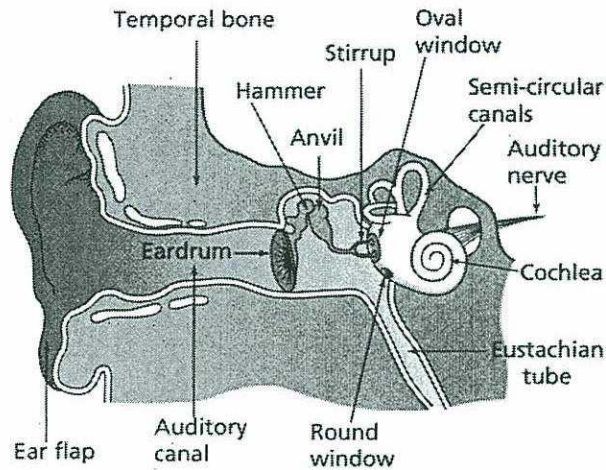
The threshold of pain is 120 dB.

Each increase by 10 dB is a ten-fold ($\times 10$) increase in the amount of sound energy being produced.



THE EAR

The ear has three major parts - the outer, middle and inner ear.



OUTER EAR

Ear flap (pinna) - collects sound waves.

Auditory canal - air vibrates here.

Eardrum - vibrates with sound waves, in contact with the hammer.

MIDDLE EAR

Hammer, anvil, stirrup - pass vibrations to the oval window.

Eustachian tube - joins middle ear to mouth, keeps pressure in the ear same as that outside.

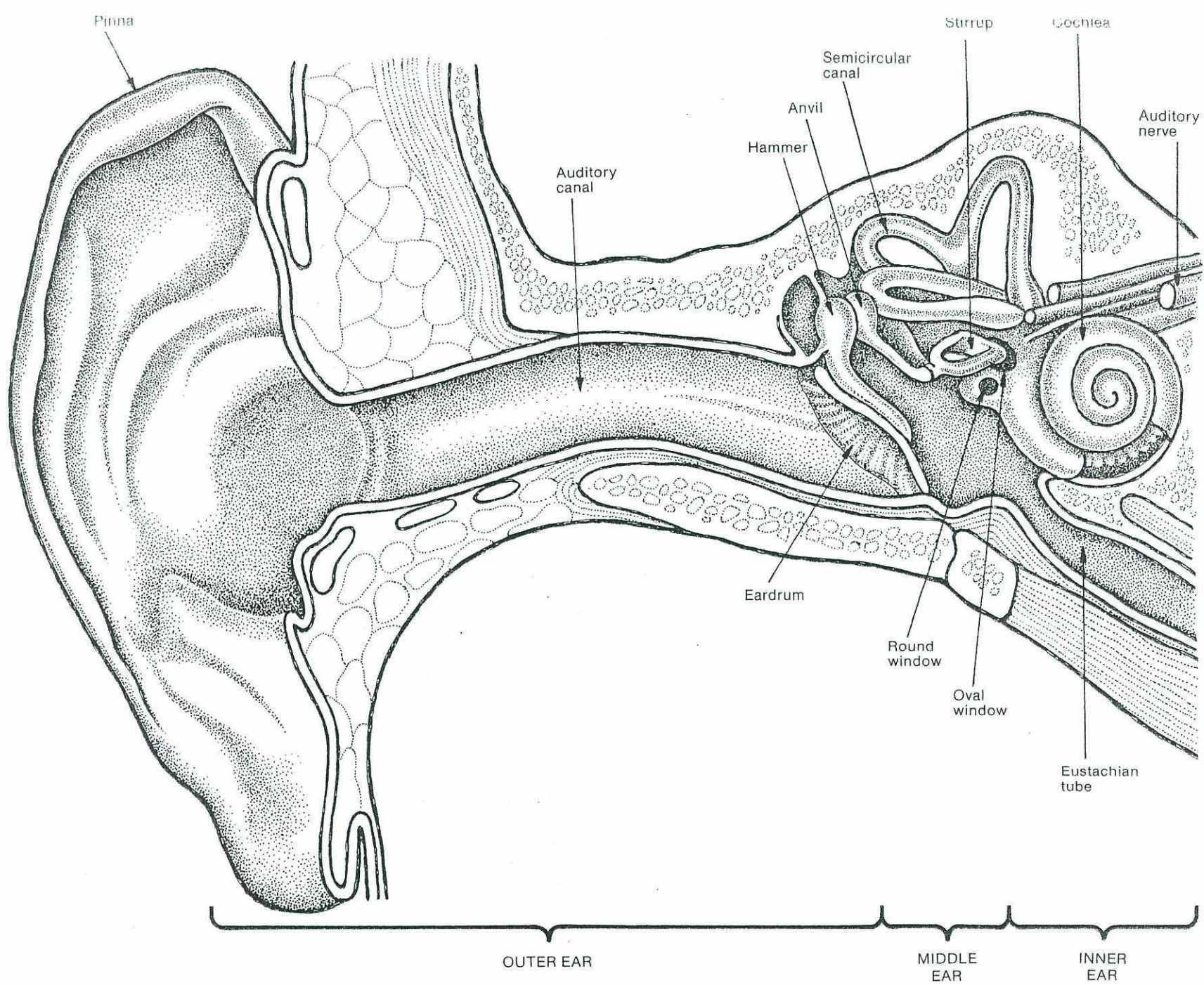
Semi-circular canals - assist in body balance.

INNER EAR

Cochlea - contains nerve endings that change vibrations into messages for the brain.

Auditory nerve - carries messages to the brain.

THE EAR



HOW DOES THE EAR WORK?

LABEL THE EAR

THINGS TO FIND OUT

WHAT DO THESE PARTS OF THE EAR DO?

Pinna _____

Auditory canal _____

Eardrum _____

Hammer, anvil, stirrup _____

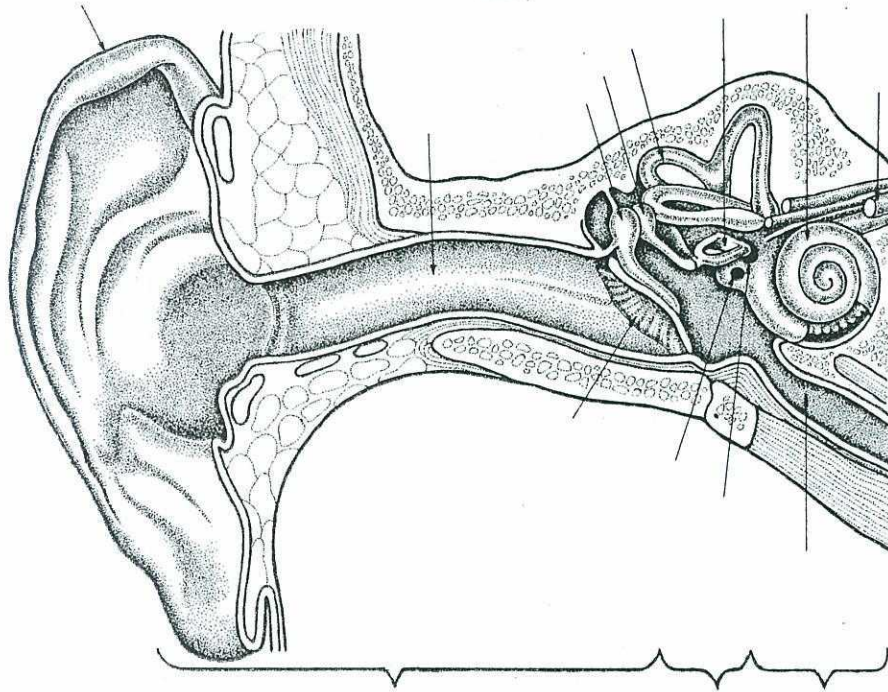
Semi-circular canals _____

Eustachian tube _____

Cochlea _____

Auditory nerve _____

Nerve cells (hair cells) _____



All the labels needed to complete the diagram have been listed. Add the labels to the diagram.

Sound is produced by vibrating objects that cause the molecules in a transmitting medium (here taken as air) to vibrate. These vibrations are called sound waves. Sound waves travel through the air from a vibrating object to our ears.

Our ears can pick up different pitches. Pitches depend on the frequency of a sound wave – a high frequency vibration makes a high-pitched note, while a low frequency vibration makes a low-pitched note. Our ears can detect sounds ranging from 20 vibrations per second to 20 000 vibrations per second.

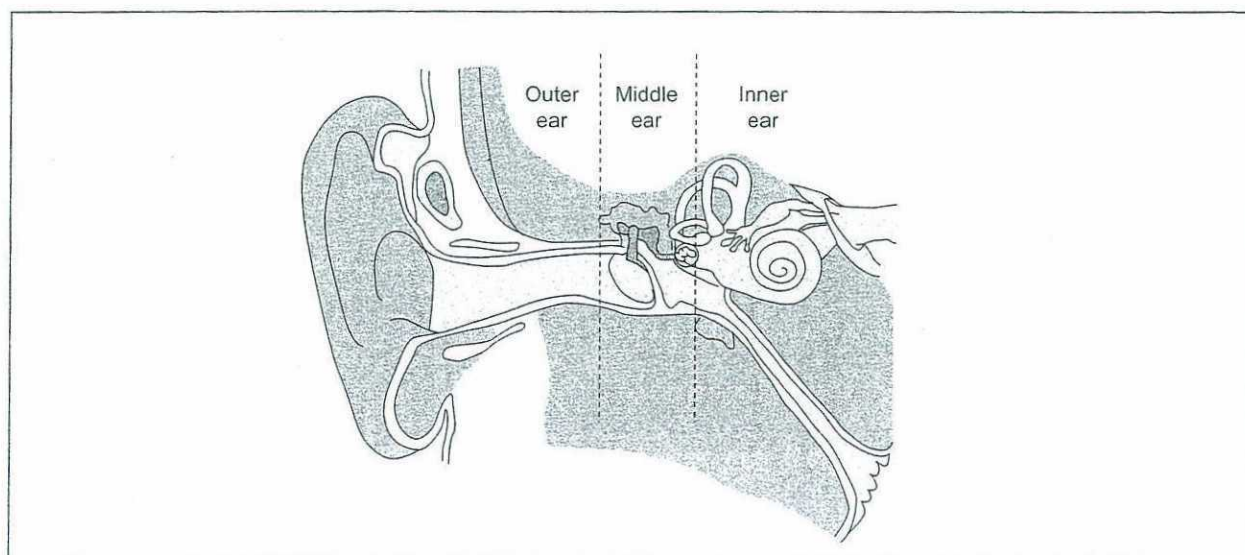
Below is a table detailing the different parts of a human ear.

Part	Location	Function
Ear shell	Outer ear	Open funnel that collects sound waves
Ear canal	Outer ear	Long canal that leads sound waves to the ear drum
Eardrum	Middle ear	Thin membrane that vibrates when sound waves come in contact with it. It moves at the same frequency as the sound waves that hit it.
Hammer	Middle ear (surrounded by air)	Small bone near the eardrum. It picks up the vibrations from the eardrum and magnifies them.
Anvil	Middle ear (surrounded by air)	Hammer sends vibrations to the anvil, which is next to it. This is also a small bone that vibrates and magnifies vibrations.
Stirrup	Middle ear (surrounded by air)	Another small bone that picks up the vibrations from the anvil and magnifies them. It then vibrates against the inner ear.
Cochlea	Inner ear (filled with fluid)	Fluid filled tube that is coiled (looks like a snail shell). It contains receptors that detect vibrations in the fluid. Receptors send signals to the brain.
Semi-circular canals	Inner ear (filled with fluid)	Three tubes filled with fluid, each tube positioned in a different plane. They help with balance. Tiny hairs in the tubes trigger receptor cells that send signals about any movements to the brain.
Auditory nerve	Inner ear	Nerve that carries signals from the cochlea to the hearing centre of the brain. The brain then interprets these as sound.

1 a Use the table above. Which part of the ear has nothing to do with hearing? _____

b Explain why this part is very important. _____

2 Use information from the table. Label the diagram of the ear.



- 3** In your science book, rewrite the information given in the table as a report. Don't forget a title and an introduction. Use the lines below to plan your report.

- 4** When you have a middle ear infection, the middle ear partly fills with a thick liquid. Explain why hearing would be difficult for you if this happened.

- 5** If there is an extremely loud noise near you, your ear may not operate properly.

a Describe what could have happened to your eardrum and explain what exactly would have caused that.

b Explain why there could be a ringing in your ears that starts in the cochlea.

- 6** A tube connects your middle ear with your mouth cavity. When you fly up in a plane or drive up a long, steep hill, your ears 'pop' as you open your mouth or swallow. Assume that air moves only slowly in that tube and explain what causes your ears to pop.

- 7** Somebody hits you on the head and you hear a ringing or buzzing noise. Your ears themselves are in good working order and there is no such noise outside. Explain where the noise may come from.

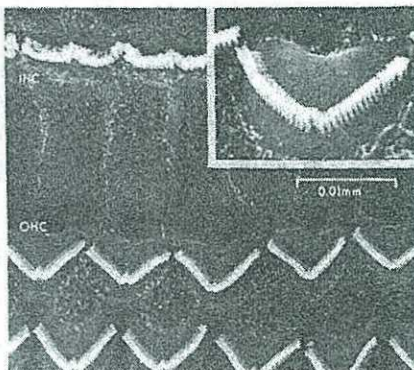
- 8** When you spin on the spot for a couple of minutes, the liquid inside the semi-circular canals spins with you. But when you suddenly stop spinning, the liquid doesn't, and you feel as if you are spinning the opposite way. Explain why you feel dizzy and find it difficult to walk properly.



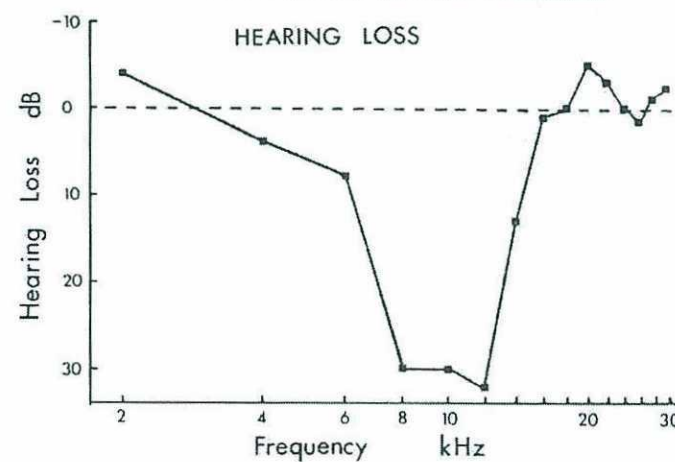
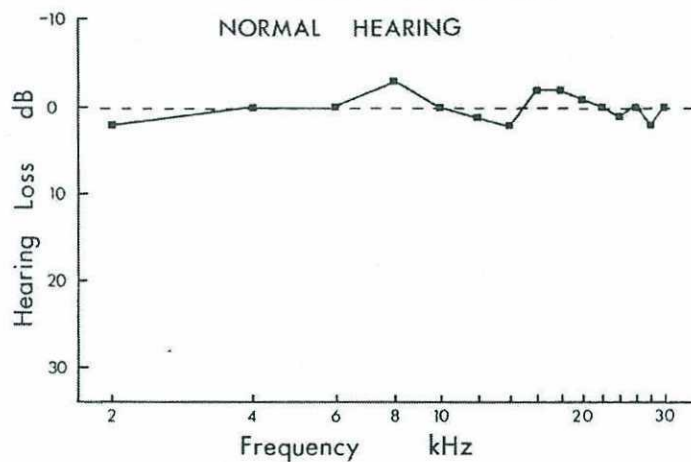
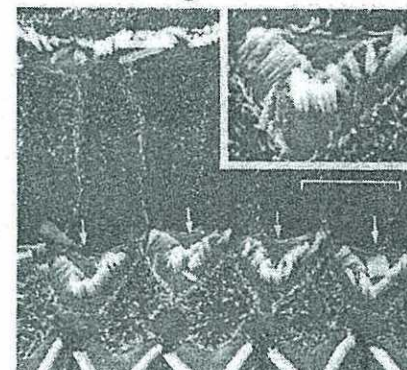
DECIBEL DANGER



Normal Nerve Cells



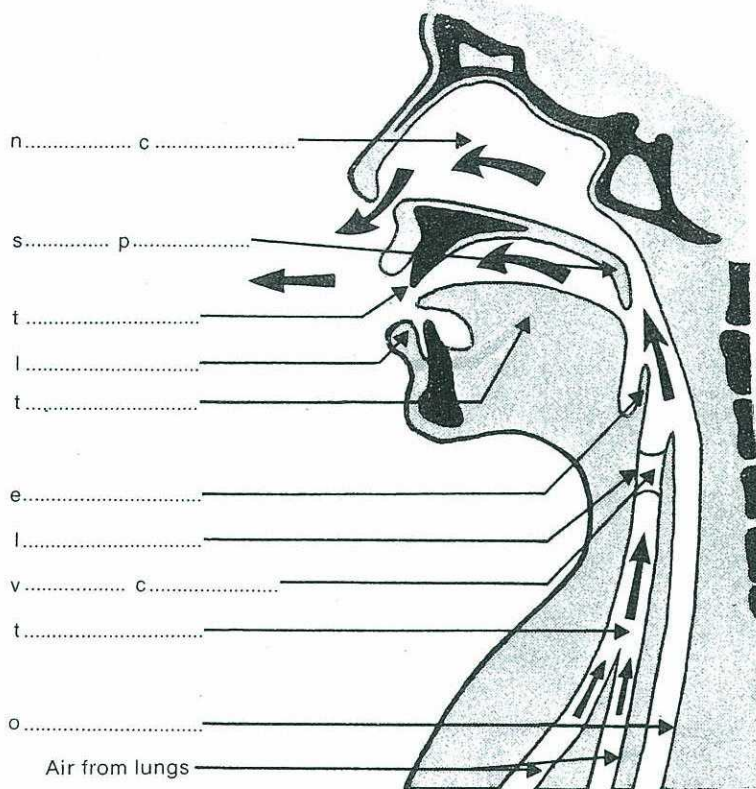
After Prolonged Loud Noise



TRY THESE

1. What are the effects on hearing and the ear of prolonged exposure to noise?
2. Examine the audiograms and explain what has happened to the person's ability to hear in the range of 4000 - 10 000 hertz?
3. From the information available on the activity sheet, are all nerve cells equally sensitive to noise damage?
4. A person with this type of disability may be able to use a hearing aid. What special features do you think the hearing aid would have?

SPEECH



Fit the labels: lip, oral cavity, epiglottis, nasal cavity, larynx, trachea, oesophagus, soft palate, tongue, teeth, vocal cords

VIEW FROM
ABOVE THE
LARYNX

Breathing

Muscle

Vocal
cords

Talking

Muscle

Vocal
cords

RESEARCH

Use a book to find out how we produce speech.

How your vocal cords work



- Inflate the balloon until it is 15 cm to 20 cm across.
- Hold onto the balloon and let the air out (deflate).
- Inflate the balloon again.
- Let the balloon deflate, but this time pinch the neck between both thumbs and forefingers. (See diagram.)
- As it deflates, move your hands apart and together.
- Describe what you heard and saw. _____
- In what ways does the balloon resemble your lungs and vocal cords? _____

